Name: Team Reef **Date:** February 3, 2026 **Class:** Engineering

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

That our ROV collects microplastic data 1-5mm. This design requirement directly to our prior research because this the average size that we found microlastcs. Our stakeholder said that this is good but it also doesn't really matter if the sizes of the microplastics go outside of this measurement.

- ☑ **Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.**
- ☑ **Explain how this design requirement directly ties to your prior research.**
- ☑ **Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.**

Design Requirement #2 (HIGH Priority)

Create a successful system that can collect and hold microplastics, and bring them back to land without dropping any microplastics back into the water. This design requirement ties directly to our prior research because we found that there would be no point if we collected microplastcis but weren't able to bring them back to shore to study them. Our stakeholder only said that this is correct to be a high priority.

- ☑ **Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.**
- ☑ **Explain how this design requirement directly ties to your prior research.**
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Design Requirement #3 (HIGH Priority)

Collects a minimum of .5-1 gram of microplastics. This design requirement ties directly to our prior research because this amount we found would be enough for a scientist to be able to study. Our stakeholder said that this amount would be sufficient.

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- ☑ **Explain how this design requirement directly ties to your prior research.**
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Design Requirement #4 (HIGH Priority)

In the process of collecting microplastics, it is ensured that the surrounding ecosystem won't be harmed or damaged in any way. This design requirement ties directly to our prior research because we found that there was a problem with ROV damaging the surround life. Our stakeholder said that this is important but not as important as collecting the microplastics themselves.

- ☑ **Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.**
- ☑ **Explain how this design requirement directly ties to your prior research.**
- ☑ **Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.**

Design Requirement #5 (HIGH Priority)

The mechanism that is used to collect the microplastics should be no smaller than 1 foot and no bigger than 4 feet. This design requirement ties directly to our prior research because we found that most other microplastic collection devices were all around this size. Our stakeholder said that this is a good size to work with but that we should rather focus on ensuring that the prototype will work and try not to worry too much about the size.

- ☑ **Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.**
- ☑ **Explain how this design requirement directly ties to your prior research.**

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Design Requirement #6 (LOW Priority)

Doesn't break, fall apart, or drown underwater. This design requirement ties directly to our prior research because durability is very important for something that has an important job such as a underwater microplastic collection device. Our stakeholder said that this is important because we don't want to end up leaving any plastics in the water.

- ☑ **Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.**
- ☑ **Explain how this design requirement directly ties to your prior research.**
- ☑ **Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.**

Design Requirement #7 (LOW Priority)

It looks nice. Isn't messy, or has useless parts attached to it. This design requirement ties directly to our prior research because we learned that if there are any unnecessary parts on our ROV, it could affect how it functions. Our stakeholder said that this is a good thing to keep in mind but that it is not as important.

- ☑ **Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.**
- ☑ **Explain how this design requirement directly ties to your prior research.**
- ☑ **Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.**